

Net Electricity Demand Forecasting in France

Statistical learning for the 2022–2023 sobriety period

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May 29, 2026

Abstract

We forecast the daily *net* electricity demand of France (consumption minus wind and solar production) over the 2022–2023 “sobriety period”. A family of statistical learners — additive models, an online state-space (Kalman) adaptation, quantile additive models and tree ensembles — is trained and combined through online expert aggregation. Models are evaluated with the pinball loss at quantile 0.8, the competition metric, which deliberately penalises under-forecasting. The Kalman-adapted GAM and the expert aggregation give the most reliable forecasts.

1 Introduction

Balancing the electricity grid requires accurate short-term demand forecasts; a mismatch between supply and demand wastes energy or risks blackouts. We study the *net demand*, the load that renewable (solar and wind) production cannot cover,

$$\text{Net_demand} = \text{Load} - (\text{Wind_power} + \text{Solar_power}),$$

during the French “sobriety period” (September 2022–September 2023), when consumption patterns shifted. Forecasts are scored with the **pinball loss at quantile $\tau = 0.8$** ,

$$\mathcal{L}_\tau(y, \hat{q}) = (\tau - \mathbf{1}\{y < \hat{q}\}) (y - \hat{q}),$$

which is asymmetric: under-forecasting the demand is penalised more heavily than over-forecasting, a conservative choice for grid security.

2 Data

Daily observations span 2013 to September 2022 for training, with the sobriety period held out for prediction. The covariates combine calendar information (time of year, day of week, public holidays, school breaks, daylight-saving), weather (temperature and smoothed/min/max variants, wind, nebulosity) and lagged production/consumption (one-day and one-week lags of load, net demand, solar and wind power). The data contains no missing values, so all variables are kept.

3 Methodology

Each model uses an estimation technique and covariate set suited to it; the final prediction combines them. While developing the models we validate on the most recent years before touching the test horizon.

GAM + Kalman filter. A Generalized Additive Model captures the smooth, non-linear effects of temperature, time of year and lagged demand. Its term contributions are then adapted online with a state-space (Kalman) filter whose variances are tuned by expectation–maximisation, letting the model track the regime change of the sobriety period. This is the single most reliable model.

Ridge GAM. A GAM with a double (ridge) penalty and automatic term selection yields a frugal, robust fit that resists over-fitting from overly flexible splines.

GAM + Random Forest on residuals. A Random Forest is fitted to the GAM residuals (on temperature, time of year, weekday and a time index) to recover non-linear structure missed by the additive model.

Random Forest. A tree ensemble over the full covariate set (`mtry` tuned), strong for capturing interactions but prone to over-smoothing trends.

Quantile GAMs. Quantile GAMs at $q = 0.2$ and $q = 0.8$ bound the predictions and directly target the asymmetric loss.

Expert aggregation. Finally, all models are combined online with the `opera` mixture (MLpol) under the pinball loss, so the ensemble weights adapt sequentially to whichever expert performs best.

4 Results

Table 1 reports validation performance (RMSE and pinball loss at 0.8). The Kalman-adapted GAM gives the best overall trade-off and the lowest RMSE, while the Random Forest reaches the lowest pinball loss; the online aggregation combines their strengths for the final submission.

Model	RMSE	Pinball@0.8
GAM + Kalman	1629	537
Ridge GAM	1353	562
GAM + RF (residuals)	1754	716
Random Forest	2064	465

Table 1: Validation performance (lower is better).

5 Conclusion

Combining additive models with online Kalman adaptation, quantile regression and tree ensembles — and aggregating them online — produces robust net-demand forecasts through a period of shifting consumption. The asymmetric pinball objective is handled explicitly by the quantile GAMs and the aggregation. Natural extensions include richer state-space dynamics and probabilistic ensembling across quantiles.

References

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